

Functional Annotation Report: *dapE* (Q88MP5, PP_1525) — Succinyl- diaminopimelate Desuccinylase in *Pseudomonas putida* KT2440

Summary

The gene **dapE** (ordered locus **PP_1525**; UniProt **Q88MP5**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125) encodes **succinyl-diaminopimelate desuccinylase (SDAP desuccinylase; EC 3.5.1.18)**, a soluble cytoplasmic metalloenzyme of the **peptidase M20A family, DapE subfamily**. Its primary and specific function is to catalyze the **hydrolytic cleavage of the amide (succinyl) bond of N-succinyl-L,L-2,6-diaminopimelate (SDAP)**, releasing **L,L-2,6-diaminopimelate (L,L-DAP) and succinate** (Rhea:22608). This is the third and final step of the **succinylase branch** of the diaminopimelate (DAP)/lysine biosynthetic pathway, and its product feeds directly into two essential downstream fates: epimerization/decarboxylation to L-lysine for protein synthesis, and conversion to meso-DAP for cross-linking of the peptidoglycan cell wall.

Identity verification was rigorous and positive. The gene symbol, EC number, family, and domain architecture assigned by UniProt/HAMAP (rule MF_01690) match the biochemically and structurally characterized DapE orthologs described throughout the literature. The *P. putida* protein is 383 residues (~41.2 kDa), shares **62% sequence identity with the biochemically characterized *Escherichia coli* DapE**, and has **100% conservation of all seven catalytic and metal-binding residues** (His73, Asp75, Asp107, Glu141, Glu142, Glu170, His356). A very-high-confidence AlphaFold model (mean pLDDT 96.4) recapitulates the canonical two-domain M20A DapE fold. No literature for a different, unrelated gene sharing the "dapE" symbol was encountered — the annotation is unambiguous.

Two nuances are important. First, **no direct enzymatic or genetic study of the *P. putida* KT2440 DapE protein itself exists** — the functional assignment is a strong bioinformatic/evolutionary inference anchored on high sequence identity, complete active-site conservation, full pathway context, and structural modeling, rather than on a dedicated in vitro assay of

Q88MP5. Second, unlike bypass-lacking pathogens (e.g., *Helicobacter pylori*, *Mycobacterium smegmatis*) where dapE deletion is lethal, *P. putida* KT2440 **also encodes a DapL-type L,L-DAP aminotransferase (PP_3786)** that converts tetrahydrodipicolinate directly to L,L-DAP in a single step, bypassing the entire succinylase branch. DapE is therefore expected to be **genetically redundant** in this organism, even though the enzyme itself is fully functional.

Key Findings

Finding 1 — DapE catalyzes hydrolytic desuccinylation of N-succinyl-L,L-DAP to L,L-DAP + succinate (EC 3.5.1.18)

The defining catalytic activity of DapE is the hydrolysis of the amide bond linking a succinyl group to the α-amino group of L,L-2,6-diaminopimelate. The reaction is:



This is annotated for Q88MP5 through HAMAP-Rule MF_01690 as EC 3.5.1.18 (N-succinyl-L,L-2,6-diaminoheptanedioate amidohydrolase). The reaction identity is not merely a database assertion: it is grounded in direct enzymatic and structural work on orthologs. Crystallographic product-bound structures of DapE unambiguously establish that the enzyme "converts N-succinyl-L,L-diaminopimelic acid to L,L-diaminopimelic acid and succinate" ([PMID: 20138056](#)). A validated ninhydrin/spectrophotometric assay quantifies the desuccinylase activity of *Haemophilus influenzae* DapE and confirms enzyme identity — "succinyl-diaminopimelate desuccinylase (DapE, E.C. 3.5.1.18)" — reporting inhibitor potencies (e.g., captopril $\text{IC}_{50} \approx 3.4 \mu\text{M}$) that match the EC number applied to Q88MP5 ([PMID: 29698518](#)). Because *P. putida* DapE shares this exact catalytic machinery (see Findings 5–6), the same reaction and substrate specificity are assigned to it.

Finding 2 — DapE is a dinuclear metallohydrolase acting as a homodimer with catalytic and dimerization domains

DapE enzymes are **dinuclear (two-metal) metallohydrolases**. High-resolution X-ray structures of *H. influenzae* DapE resolved "one and two zinc ions bound in the active site" and demonstrated distinct catalytic behavior for the mono- versus di-metalated forms (PMID: 20138056). Each subunit "consisted of a catalytic domain and a dimerization domain" (PMID: 25204745). A striking, mechanistically central feature is a **large substrate-induced conformational change**: upon substrate/product binding, "the catalytic domain rotates $\sim 50^\circ$ and shifts ~ 10.1 Å (as measured at the position of the Zn atoms) relative to the dimerization domain," closing over the dinuclear Zn(II) cluster (Zn–Zn distance ~ 3.8 Å) and completing the oxyanion hole — notably with His194 contributed *in trans* by the opposing subunit (PMID: 29272107). This inter-subunit contribution explains why the **homodimer** is the functional unit. Spectroscopic studies of Co(II)-substituted enzyme show the two metals are antiferromagnetically coupled and can be bridged by a thiolate when the inhibitor captopril binds (PMID: 26186504).

Finding 3 — DapE operates in the succinylase branch of the DAP/lysine pathway, supplying peptidoglycan and lysine precursors

DapE acts at a **late stage of the anabolic pathway** leading to both L-lysine and meso-DAP, catalyzing "the desuccinylation of N-succinyl-L,L-diaminopimelic acid (SDAP) in a late stage of the anabolic pathway towards both lysine and a crucial building block of the peptidoglycan cell wall" (PMID: 24057071). Genetic ordering studies in *E. coli* (using ΔdapD ΔdapE mutants) place dapE between the succinyl-transfer/aminotransferase steps and the DapF epimerase (PMID: 17093042). The biological importance of this reaction is that its downstream products are "essential for protein synthesis and the cross-linking of the bacterial peptidoglycan cell wall" (PMID: 40973684). Thus DapE sits at a metabolic branch point that provisions two indispensable macromolecular processes.

Finding 4 — DapE is essential in many bacteria, has no human homolog, and is a validated antibacterial target

Across bypass-lacking bacteria, dapE is **essential**: its deletion is lethal to *H. pylori* and *M. smegmatis* (PMID: 20138056), and lethal to *H. pylori* specifically because "deletion of DapE gene is lethal to *H. pylori*, since the organism has no alternative pathway for lysine biosynthesis" (PMID: 25204745). Because "there are no similar pathways in humans, inhibitors that target

DapE may have selective toxicity against only bacteria" (PMID: 20138056). DapE is highly conserved across diverse bacterial species (PMID: 40973684). One important pharmacological caveat is **metal heterogeneity**: the physiologically relevant "Mn(2+)-enzyme, which is also a physiologically relevant isoform in bacteria, is not inhibited" by L-captopril, which targets only the Zn(2+) isoform (PMID: 24057071). This finding contextualizes DapE as a drug target in general; the redundancy in *P. putida* (Finding 7) means DapE is unlikely to be an effective target in this non-pathogenic organism specifically.

Finding 5 — *P. putida* DapE has a fully conserved dinuclear metal active site with residue-level mapping

The 383-residue Q88MP5 sequence (41.2 kDa) carries residue-level HAMAP annotations that reproduce the canonical M20A dinuclear coordination sphere. The **active sites are Asp75 and Glu141** (the latter acting as the proton acceptor / catalytic general base), and the **metal ligands are His73, Asp107 (bridging both metals), Glu142, Glu170, and His356, binding two Zn(2+) or Co(2+) ions per subunit**. Direct mapping onto the primary sequence confirms these residues and their local motifs (...FAGH-TDV... at positions 73–75; ...TSD-EE... at 141–142), matching the classic "2 His + bridging Asp + Glu general base" arrangement of the peptidase M20A dinuclear metallopeptidases. Supporting cross-references include Rhea:22608 for the catalytic activity, pathway UPA00034/UER00021 (L-lysine biosynthesis via DAP, succinylase route, step 3 of 3), homodimer quaternary structure, family peptidase M20A/DapE subfamily, eggNOG COG0624, CDD cd03891 (M20_DapE_proteobac), and TIGRFAM TIGR01246 (dapE_proteo). These conserved residues are interpreted against the canonical DapE mechanism, for which the ortholog structural work proposed "a revised catalytic mechanism of peptide bond cleavage by DapE enzymes" (PMID: 20138056).

Finding 6 — 62% identity to characterized *E. coli* DapE with 100% catalytic-residue conservation; gene is monocistronic

A global Needleman–Wunsch alignment of Q88MP5 (383 aa) against biochemically characterized *E. coli* DapE (P0AED7, 375 aa) yields **232/374 identical residues = 62.0% identity over the full length**. Critically, **all seven functional residues are strictly conserved**: His73, Asp75, Asp107, Glu141, Glu142, Glu170, and His356 each align to the identical residue in *E. coli*. KEGG independently annotates PP_1525 as **K01439** (succinyl-diaminopimelate desuccinylase, EC 3.5.1.18) in pathway ppu00300 (Lysine biosynthesis). Genomic context analysis shows PP_1525 occupies complement(1729936..1731087), flanked by PP_1524 (23S rRNA methyltransferase), PP_1526 (glucosyltransferase), PP_1523/PP_1528 (unknown), and PP_1522

(cold-shock protein) — **none are other DAP/lysine pathway genes** (dapA/B/C/D/E, lysA, asd map elsewhere on the chromosome). The gene is therefore **effectively monocistronic and independently regulated**, not co-transcribed in a dap operon. The shared function of the DapE orthologs to which the 62%-identical *P. putida* protein is compared is captured by the observation that "the dapE operon enzymes synthesize both meso-diaminopimelic acid and lysine" ([PMID: 20138056](#)).

Feature	<i>P. putida</i> DapE (Q88MP5)	<i>E. coli</i> DapE (P0AED7)
Length	383 aa (~41.2 kDa)	375 aa
Sequence identity	—	62.0% (232/374)
His73 / Asp75 / Asp107	conserved	identical
Glu141 / Glu142 / Glu170	conserved	identical
His356	conserved	identical
Metal ions per subunit	2 (Zn ²⁺ or Co ²⁺)	2 (Zn ²⁺)
KEGG ortholog	K01439	K01439
Genomic context	monocistronic (PP_1525)	—

Finding 7 — *P. putida* encodes both the succinylase branch and a DapL bypass, so dapE is likely genetically redundant

The KEGG ortholog inventory for *P. putida* KT2440 (ppu) confirms the **succinylase branch is present**: dapD = PP_1530 (K00674); the N-succinyl-DAP aminotransferase (dapC) step is supplied by the promiscuous bifunctional **ArgD = PP_0372** (K00821; a dedicated dapC/K05822 gene is absent); and dapE = PP_1525 (K01439). Downstream steps are present: dapF = PP_3790 (K01778) and lysA = PP_2077/PP_5227 (K01586). Critically, *P. putida* **also encodes a DapL-type L,L-diaminopimelate aminotransferase, PP_3786** (K10206, EC 2.6.1.83) — "an aminotransferase that may directly convert tetrahydrodipicolinate to L,L-diaminopimelic acid" ([PMID: 17093042](#)) — bypassing the entire dapD→dapC→dapE succinylase branch in a single transamination. The meso-DAP dehydrogenase (ddh, K03340) and acetylase-branch genes (dapH/dapX) are absent. This contrasts sharply with organisms lacking any bypass, where "deletion of DapE gene is lethal to *H. pylori*, since the organism has no alternative pathway for lysine biosynthesis" ([PMID: 25204745](#)). Hydropathy analysis of Q88MP5 shows **no**

transmembrane segment (max Kyte–Doolittle 19-residue window = 0.85; 0 windows above the ~1.6 threshold) and **no cleavable N-terminal signal peptide**, consistent with a **soluble cytoplasmic** localization.

Finding 8 — AlphaFold model is very high confidence, supporting the two-domain M20A DapE fold

The AlphaFold DB model for Q88MP5 (383 residues, version 6) has a **mean pLDDT of 96.4**, with **94.5% of residues at very-high confidence (pLDDT > 90)**, 98.7% confident (pLDDT > 70), and only 0.8% at low confidence (pLDDT < 50). The uniformly high confidence across the full length is consistent with a well-defined **two-domain fold** — a catalytic Peptidase_M20 domain plus an M20_dimer dimerization domain — matching the architecture observed in experimentally solved DapE structures, in which "the catalytic domain rotates ~50° and shifts ~10.1 Å ... relative to the dimerization domain" ([PMID: 29272107](#)).

Mechanistic Model / Interpretation

Position in the diaminopimelate/lysine biosynthetic pathway

DapE catalyzes step 3 of 3 in the **succinylase branch** that converts tetrahydrodipicolinate (THDP) into L,L-DAP:

Catalytic mechanism at the dinuclear metal center

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requirement for the homodimer. This mechanism, worked out from mono- and di-metalated *H. influenzae* structures and from substrate-bound conformational-change structures, applies directly to the *P. putida* enzyme because every ligand and catalytic residue is conserved.

Localization

All evidence points to a **cytoplasmic** enzyme: no transmembrane helices, no signal peptide, a soluble globular two-domain fold, and a substrate (SDAP) that is a cytoplasmic metabolic intermediate. There is no evidence for periplasmic, membrane, or secreted localization.

Reconciling function with redundancy

The key interpretive point for *P. putida* is that **an enzyme can be fully functional yet genetically redundant**. Q88MP5's active site is 100% conserved and its fold is textbook-perfect, so the protein almost certainly retains full desuccinylase activity. However, the co-existence of the DapL bypass (PP_3786) means that loss of *dapE* need not be lethal — in contrast to bypass-lacking pathogens. This distinction matters for any consideration of DapE as an antibacterial target: it is validated in pathogens with an obligate succinylase branch, but not in organisms like *P. putida* that carry the DapL shortcut.

Evidence Base

PMID	Title (abbrev.)	How it supports / challenges the findings
20138056	Structural basis for catalysis by mono-/dimetalated DapE	Defines substrate→products (L,L-DAP + succinate), dinuclear Zn active site, essentiality in <i>H. pylori</i> / <i>M. smegmatis</i> , no human homolog, revised mechanism. Core anchor for Findings 1, 2, 4, 5.
29272107	Major conformational change on substrate binding in DapE	Documents ~50° / 10.1 Å catalytic-domain closure and two-domain architecture; underpins mechanism and AlphaFold fold interpretation (Findings 2, 8).
25204745	In silico inhibitors of <i>H. pylori</i> DapE	Confirms catalytic + dimerization domain organization; establishes essentiality where no bypass exists — the key contrast for <i>P. putida</i> redundancy (Findings 2, 4, 7).
24057071	Zinc-selective inhibition / metal heterogeneity	Places DapE in late DAP/lysine pathway toward lysine + peptidoglycan; documents Mn ²⁺ vs Zn ²⁺ isoform inhibitor differences (Findings 3, 4).
29698518	Spectrophotometric DapE assay	Validated activity assay confirming enzyme identity/EC 3.5.1.18 and inhibitor IC ₅₀ values (Finding 1).
26186504	L-captopril inhibition of <i>N. meningitidis</i> DapE	EPR/MCD/crystallography showing antiferromagnetically coupled dinuclear metals and thiolate bridge on inhibitor binding (Finding 2).
17093042	L,L-DAP aminotransferase (DapL) trans-kingdom	Establishes the DapL bypass that directly converts THDP to L,L-DAP, the basis for <i>P. putida</i> redundancy; helps order the succinylase branch (Findings 3, 7).
40973684	Zinc-binding DapE inhibitors in <i>K. pneumoniae</i>	States biological role (protein synthesis + peptidoglycan cross-linking) and broad conservation of DapE (Findings 3, 4).
32860896	DapE–flavonoid interactions	Reinforces DapE as an antimicrobial metal-center target across genera including <i>Pseudomonas</i> ; supports the general drug-target framing.

Database evidence: UniProt/HAMAP MF_01690 (EC 3.5.1.18; residue-level active-site and metal annotations), Rhea:22608, MetaCyc/UniPathway UPA00034/UER00021, KEGG K01439 / ppu00300 / PP_1525, eggNOG COG0624, CDD cd03891, TIGRFAM TIGR01246, and AlphaFold DB (mean pLDDT 96.4). These are consistent with, and mutually reinforced by, the primary literature above.

Limitations and Knowledge Gaps

1. **No direct study of the *P. putida* KT2440 protein.** Every functional claim about Q88MP5 is a bioinformatic/evolutionary inference. No purified-protein kinetics, metal-content determination, crystal structure, or gene-knockout phenotype exists for this specific ortholog. The inference is strong (62% identity to a characterized enzyme, 100% active-site conservation, complete pathway context, pLDDT 96.4) but remains inference.
 2. **Redundancy is predicted, not demonstrated.** The expectation that dapE is dispensable in *P. putida* rests on the presence of the DapL gene (PP_3786) and comparison to bypass-lacking organisms. No knockout has been reported to confirm that Δ dapE is viable in *P. putida*, nor whether DapL flux is sufficient under all growth conditions. It is possible the two routes are differentially regulated or condition-dependent.
 3. **Physiological metal identity is unresolved.** DapE orthologs show metal heterogeneity (Zn^{2+} vs. Mn^{2+} vs. Co^{2+}) with functional and pharmacological consequences. The in vivo metal used by *P. putida* DapE is unknown.
 4. **Substrate stereospecificity assumed by homology.** The L,L-DAP (not meso- or D,D-) stereospecificity is inferred from orthologs and the conserved active site, not measured for Q88MP5.
 5. **Regulation is uncharacterized.** The monocistronic genomic context implies independent regulation, but no promoter, transcription-factor, or expression data were established for PP_1525 in this investigation.
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Proposed Follow-up Experiments / Actions

1. **Recombinant expression and kinetic assay.** Clone PP_1525, purify Q88MP5, and measure desuccinylase activity on synthetic SDAP using the validated spectrophotometric/ninhydrin assay ([PMID: 29698518](#)). Determine k_{cat} , K_m , and metal dependence (apo-reconstitution with Zn^{2+} , Mn^{2+} , Co^{2+}).
 2. **Gene knockout / bypass test.** Construct a $\Delta dapE$ (ΔPP_{1525}) mutant and test viability and lysine/DAP auxotrophy with and without a co-deletion of the DapL gene (ΔPP_{3786}). A synthetic-lethal $\Delta dapE \Delta dapL$ phenotype rescued by exogenous lysine/DAP would directly confirm both DapE function and the redundancy model.
 3. **Structural confirmation.** Solve the X-ray or cryo-EM structure of Q88MP5, ideally with substrate/product or captopril bound, to confirm the dinuclear metal geometry and the substrate-induced domain closure predicted by the AlphaFold model and ortholog structures.
 4. **Metal-content analysis.** Use ICP-MS/EPR on natively purified enzyme to determine the physiologically bound metal(s), informing both mechanism and any inhibitor-design considerations.
 5. **Expression profiling.** Quantify PP_1525 transcript/protein levels across growth phases and nitrogen/carbon conditions to test the independent-regulation hypothesis implied by its monocistronic context, and compare against DapL (PP_3786) expression to assess which route dominates flux.
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Conclusion

dapE (Q88MP5 / PP_1525) in *Pseudomonas putida* KT2440 is confidently annotated as **succinyl-diaminopimelate desuccinylase (EC 3.5.1.18)**, a soluble cytoplasmic homodimeric dinuclear-metal hydrolase of the peptidase M20A / DapE subfamily. It hydrolyzes N-succinyl-L,L-2,6-diaminopimelate to L,L-2,6-diaminopimelate plus succinate, performing the final L,L-DAP-forming step of the succinylase branch of DAP/lysine biosynthesis and thereby supplying meso-DAP for peptidoglycan cross-linking and lysine for protein synthesis. The assignment is inferential for this specific organism but strongly supported by 62% identity to characterized *E. coli* DapE with fully conserved catalytic residues, complete pathway context, and a very-high-

confidence structural model. Because *P. putida* additionally encodes a DapL aminotransferase bypass, DapE is likely genetically redundant in this organism — a notable contrast to bypass-lacking pathogens in which dapE is essential and a validated drug target.

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